Dr. Rongxin Zhu

rxzhu [at] ieee [dot] org · Google Scholar · GitHub · Academic Homepage

Biography

Dr. Rongxin Zhu received his Ph.D. in Cyberspace Security in 2024. He is currently a lecturer at **Hainan University** and an adjunct research fellow at the **Paradise Lab, University of Ottawa**.

His research focuses on underwater networks and communications, secure and trustworthy networking, embodied and semantic communications, and AI-enabled underwater perception.

Research Interests

- Underwater Networks and Communications
- Secure and Trustworthy Networking
- Embodied and Semantic Communications
- AI-Enabled Underwater Perception

Education and Academic Qualifications

- **Hainan University**, Ph.D. in Cyberspace Security
  School of Cyberspace Security, Sep. 2021 – Jun. 2024
- **University of Ottawa**, Visiting Scholar
  School of Electrical Engineering and Computer Science, Dec. 2022 – Dec. 2023
- **Nanjing University**, M.Eng. in Software Engineering
  Institute of Software Engineering, Sep. 2015 – Jun. 2017
- **Nanjing Institute of Technology**, B.S. in Computer Science and Technology
  School of Computer Science and Technology, Sep. 2011 – Jun. 2015

Selected Publications

1. **R. Zhu**, L. Sheng, K. Wu, A. Boukerche, L. Long, and Q. Yang, "Toward Efficient Underwater Visual Perception through Image Enhancement, Compression, and Understanding," *ACM Computing Surveys*, vol. 58, no. 8, Art. no. 205, pp. 1–46, 2026. DOI
   Comprehensive survey of underwater visual enhancement, compression, and understanding.
2. **R. Zhu**, A. Boukerche, L. Long, and Q. Yang, "Design Guidelines on Trust Management for Underwater Wireless Sensor Networks," *IEEE Communications Surveys & Tutorials*, vol. 26, no. 4, pp. 2547–2576, 2024. DOI
   Systematic design guidelines for trust management in underwater networks.
3. **R. Zhu**, W. Li, A. Boukerche, and Q. Yang, "Energy-Aware DRL-Based Dual-Perception Fountain Codes for Resource-Constrained UASNs," *IEEE Transactions on Sustainable Computing*, vol. 11, no. 2, pp. 111–122, 2026. DOI
   Deep-reinforcement-learning-enabled reliable coding for resource-constrained underwater networks.
4. **R. Zhu**, D. Qin, A. Boukerche, and Q. Yang, "PIA-GAN: A Physics-Informed Attention Generative Adversarial Network for Underwater Acoustic Channel Simulation," *Computer Networks*, vol. 286, Art. no. 112397, 2026. DOI
   Physics-informed generative modeling of underwater acoustic channels.
5. **R. Zhu**, A. Boukerche, and Q. Yang, "An Efficient Secure and Adaptive Routing Protocol Based on GMM-HMM-LSTM for Internet of Underwater Things," *IEEE Internet of Things Journal*, vol. 11, no. 9, pp. 16491–16504, 2024. DOI
   Learning-based secure and adaptive routing for the Internet of Underwater Things.
6. **R. Zhu**, A. Boukerche, and Q. Yang, "An Interference-Aware and Collision-Free MAC Protocol for Underwater Wireless Sensor Networks," *ACM Transactions on Sensor Networks*, vol. 21, no. 3, pp. 1–26, 2025. DOI
   Interference-aware medium access control for large-scale underwater deployments.

7. **R. Zhu**, Q. Jiang, X. Huang, D. Li, and Q. Yang, "A Reinforcement-Learning-Based Opportunistic Routing Protocol for Energy-Efficient and Void-Avoided UASNs," *IEEE Sensors Journal*, vol. 22, no. 13, pp. 13589–13601, 2022. DOI
   Reinforcement-learning-based opportunistic routing with energy and void avoidance objectives.
8. **R. Zhu**, A. Boukerche, L. Feng, and Q. Yang, "A Trust Management-Based Secure Routing Protocol with AUV-Aided Path Repairing for Underwater Acoustic Sensor Networks," *Ad Hoc Networks*, vol. 149, Art. no. 103212, 2023. DOI
   Secure routing with trust management and autonomous-underwater-vehicle-assisted path repair.
9. **R. Zhu**, A. Boukerche, P. Li, and Q. Yang, "A Resilient and Energy-efficient Source Location Privacy for Sustainable Underwater Acoustic Sensor Networks," *IEEE Transactions on Sustainable Computing*, early access, 2026. DOI
   Source-location privacy that accounts for unreliable acoustic links and constrained sensor energy.
10. **R. Zhu**, H. Zhang, P. Li, Q. Yang, and X. Huang, "A Mobility-Aware Conflict-Graph Scheduling Framework for AUV-Assisted Collaborative Underwater Acoustic Networks," *IEEE Internet of Things Journal*, vol. 13, no. 19, pp. 44749–44760, 2026. DOI
    Mobility-aware scheduling for collaborative underwater networks assisted by autonomous underwater vehicles.

Teaching

- **Python Programming**
- **Data Structures and Programming**
- **Research Methodology** — Graduate course

My teaching interests include programming fundamentals, data structures, research methods, network systems, and the responsible application of machine learning.

Academic and Editorial Services

- **Youth Associate Editor**, *Exploration*, since April 2025
- **Guest Editor**, *Network*, Special Issue on AI-Driven Maritime–Aerial Heterogeneous Fusion Networks for Next-Generation Communications, manuscript submission deadline: December 31, 2026
- **Guest Editor**, *Photonics*, Special Issue on Applications and Development of Optical Fiber Sensors, February 2025

Reviewing Services

- IEEE INFOCOM, 2024–Present
- *IEEE Internet of Things Journal*, 2023–Present
- *IEEE Transactions on Mobile Computing*, 2022–Present
- *IEEE Sensors Journal*, 2022–Present
- *IEEE Network*, 2024–Present
- *Cluster Computing*, 2024–Present
- *Wireless Personal Communications*, 2024–Present
- *Artificial Intelligence Review*, 2023–2024
- *Computer Networks*, 2022–2023